Module-1

1. Define power electronics. Mention the different power electronic circuit
2. Explain the gate characteristics of the SCR and Turnoff characteristics of SCR
3. With the help of static V-I characteristics, explain the three modes of operation of the SCR.
4. Explain class A & class B commutation with necessary circuit diagram and waveform.
5. R and RC triggering

Module-2

1. Explain the effect of free wheeling diode used in controlled rectifiers
2. Explain the operation of step up chopper (boost converter)
3. Explain the operation of step down chopper (buck converter)
4. With necessary circuit diagram and waveforms, explain the working of single phase half wave converter with R Load & RL load.
5. With necessary circuit diagram & waveforms explain the working of single phase full wave converter with R load & RL load.

Module-3

1. Define the terms (i) instrument (ii) Absolute error (iii) Relative errors (iv) Accuracy (v) Precision (vi) Resolution.
2. Explain the working of single bridge inverter with necessary circuit diagram and waveforms.
3. What is error? Explain different types of errors
4. Explain the working of isolated forward SMPS with necessary circuit diagram.

Module-4

1. With the aid of diagram explain the working of unbalanced wheatstone bridge and derive for a galvanometer current expression
2. Discuss the operation of dual slope integrating type DVM with the help of block diagram.
3. Discuss the operation of ramp type DVM with the help of block diagram. And mention the difference b/w ramp type DVM & dual slope integrating type DVM.
4. Explain the working of a digital frequence meter
5. Explain the principle of operation of digital time measurement

Module-5

1. Explain with neat diagram the PLC structure & working.
2. Describe the operation of resistive tranducer with constructional diagram & typical circuit used.
3. Explain instrumentation Amplifier using transducer bridge with the help of ckt diagram.
4. Explain the construction, working principle & operation of LVDT. show the characteristics curve.
5. Derive an expression for the gauage factor of bonded resistance wire stain gauge.

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